

The Problem at Hand

Why This Project Exists

I started this project because I kept running into the same problems with vocal rider plugins.

1. Transient Handling

[Insert examples]

A lot of riders react badly to short peaks, ducking too hard and sometimes cutting into the sound that comes right after

2. Inconsistency

A setting can work great on one part of a vocal, then completely fall apart on another, with very little indication of why

3. Too Many Controls

A lot of plugins throw a bunch of parameters at you without making it clear what the rider is actually listening for or how those controls affect its decisions

I wanted to see if I could build something that feels more predictable, easier to understand, and still capable of competing with the better commercial options out there.

To do so, I realized one way of doing a decent job at it, would be splitting vocals into categorized segments that can each be handled appropriately. And so, the idea for a vocal segment detector began.

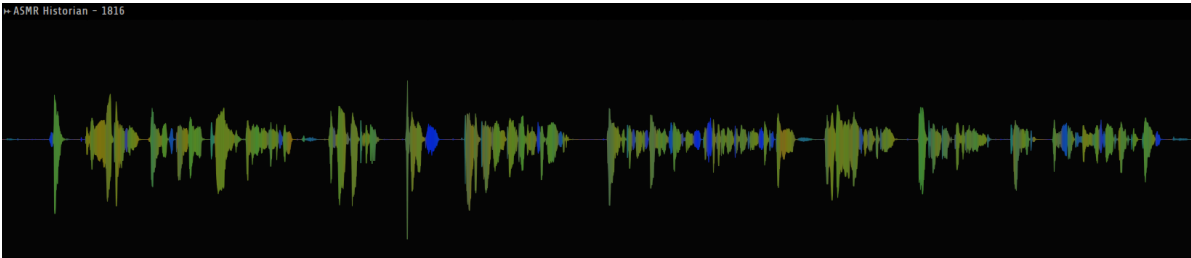
I started by looking at vocal waveforms, breaking them into different types of sounds, and figuring out what metrics can be used to indicate if a sound is a breath, plosive, sibilant, voiced sound, or transition different from one another.

Visual Demonstration

VSD'S
PROCESS

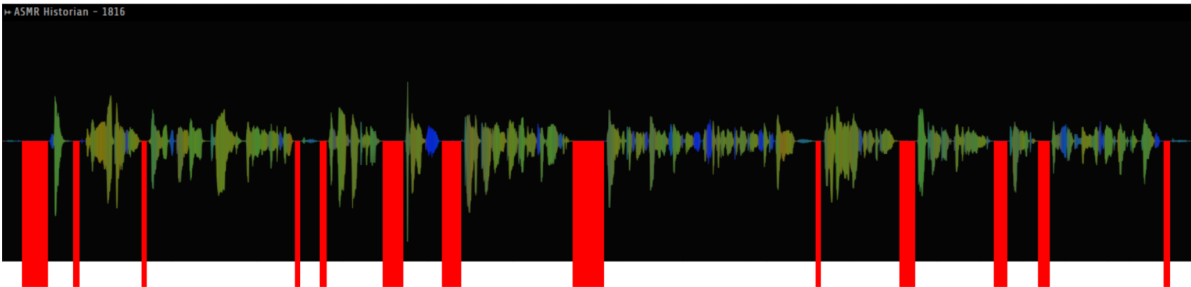
[9:27 - 9:51][ST1] "1816: The Year Without a Summer | Human Voiced,
No Ads" - ASMR Historian

SAMPLE WAVEFORM AND TRANSCRIPT



"So the mechanism by which Tambora cooled the northern hemisphere is worth talking about because it's kind of counterintuitive in its operation and because understanding it shows the specific events of 1816 and the broader relationship between volcanic activity and climate that has shaped human history at multiple points."

SIGNAL DETECTION



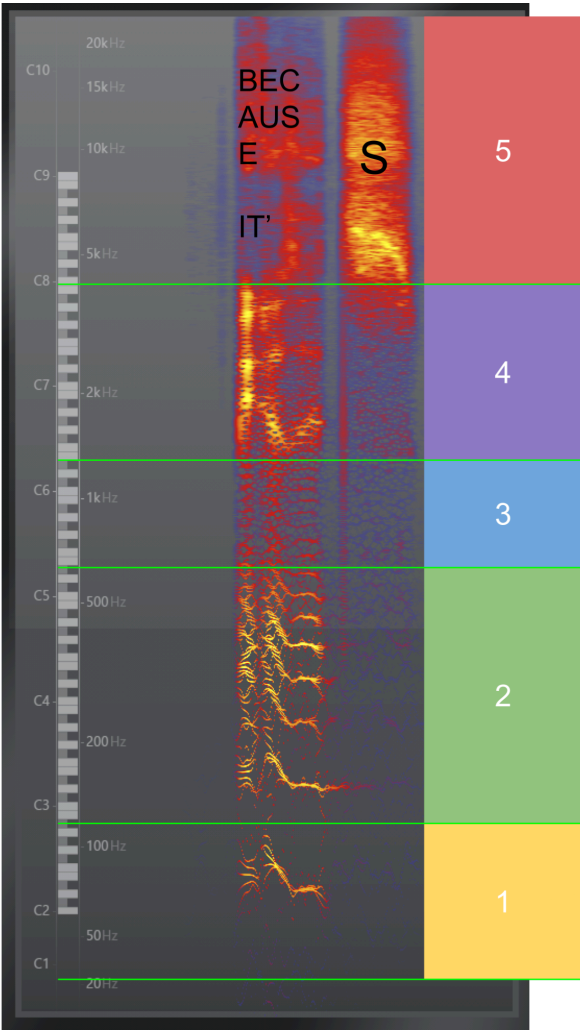
*Red areas do not meet DB Threshold

Implementation	Boolean → True if <code>inputSignal > dbThreshold</code>
Purpose	Leaves audio analysis to apply relevantly
Improvement	N/A

CATEGORIES - [ST1] excerpt

"... Because it's ..."

Band Split



See Appendix for raw spectrogram

Using the spectrogram for this snippet, green lines were drawn to split the frequency dominated spaces in the frequency range into sections.

These sections highlight some of the differences seen across a vocal. Such distinctions should allow for categorizations to be made.

Depending on the metrics a given section's, it will be placed under:

- a. Silence/noise
- b. Breath
- c. Sibilance
- d. Plosive
- e. Voiced
- f. Transition

Silence	Breath	Sibilance	Plosive	Voice	Transition
Alternatively, noise	inhales, exhales	s, sh, z, f, th	p, b, t, d, k, g		[may not be needed]
	airy "h"-type material	high-frequency consonant content	short burst-like sounds/events	vowels and nasal consonants	

The Metrics

1. Level / RMS	2. Frequency Balance	3. Harmonicity	4. Temporal Change
How much energy is present in a frame	Where energy sits across low, mid, and high frequencies	How harmonically structured the signal is	The degree and rate of change in spectrum and level
Silence	Sibilance	Voiced sounds	Plosives
Breaths	Breath	Sibilance	Transitions
Level changes	Voiced sounds	Plosives	Attacks
	Plosives		

Frame

A A frame is a structure that will hold

Samples	
Metric 1	
Metric 2	
Metric 3	
Category	

B

Appendix

Raw Spectrogram

